

HOVERR Operating Guide for Permanent Magnet Variable Frequency Air Compressor

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2025 edition

Summary of Structural Advantages of Permanent Magnet Variable Frequency Integrated Units

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Since the development and production of the post-treatment system



永磁变频电机，主机一体集成，维修，维护便捷。
内部管道设计简洁，最大可能的减少了漏油，漏气点。

Permanent magnet variable-frequency motor integrated with the main unit



永磁变频电机一体主机

The rear box panel is fully detachable, making maintenance and servicing very convenient.



背面整体箱板可拆卸，便于维护保养！

Permanent magnet variable-frequency motor, integrated with the main unit. Easy for maintenance and repair.
The internal piping design is simple, minimizing the possibility of oil and gas leakage as much as possible.



右侧大通道进风口，配带初级滤网

The large air inlet on the right side is equipped with a primary dust filter screen.

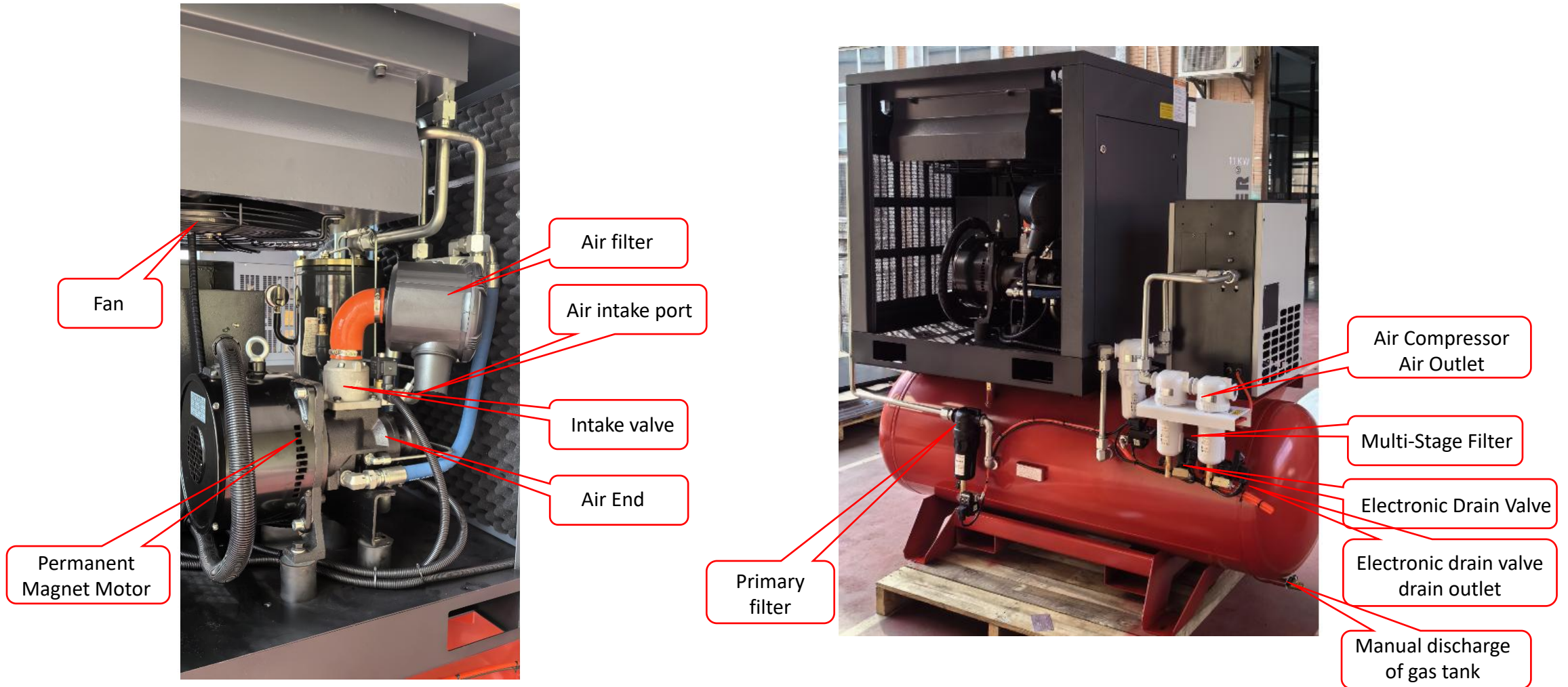
Example of Structure Introduction (Front)

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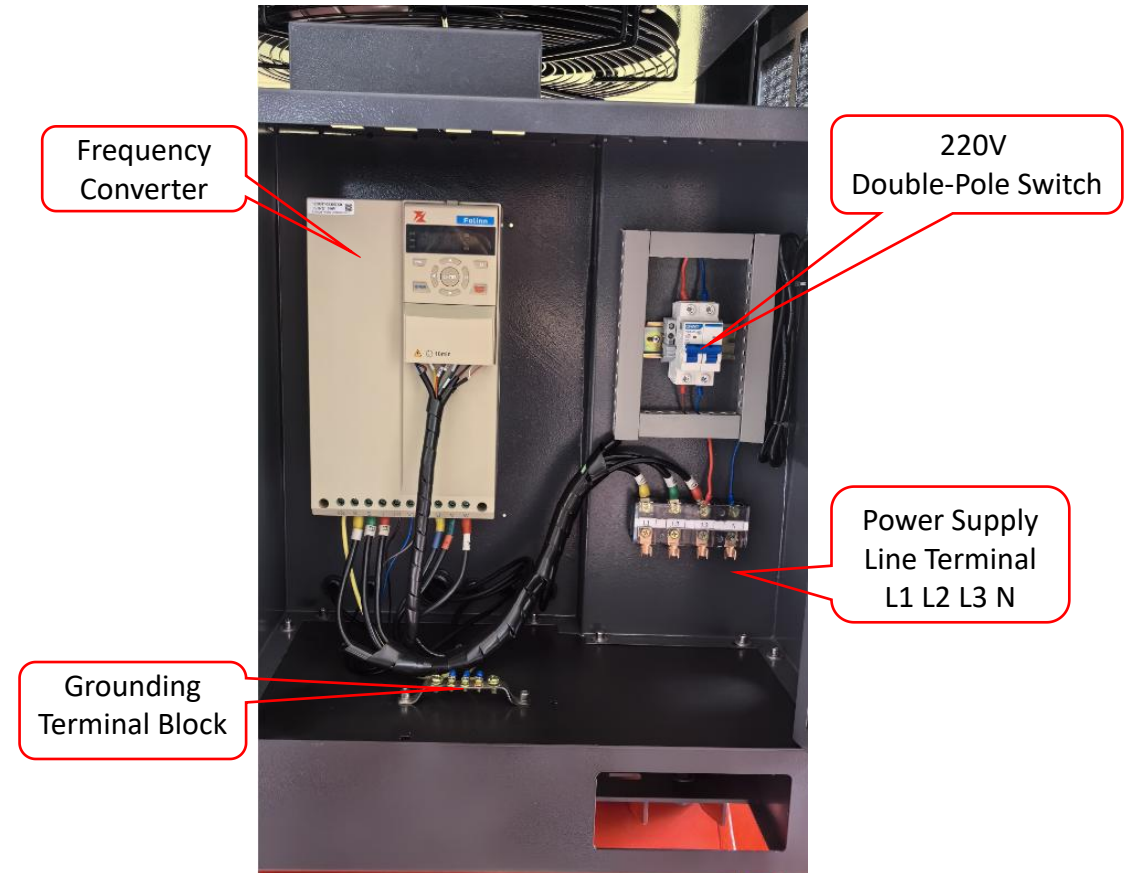
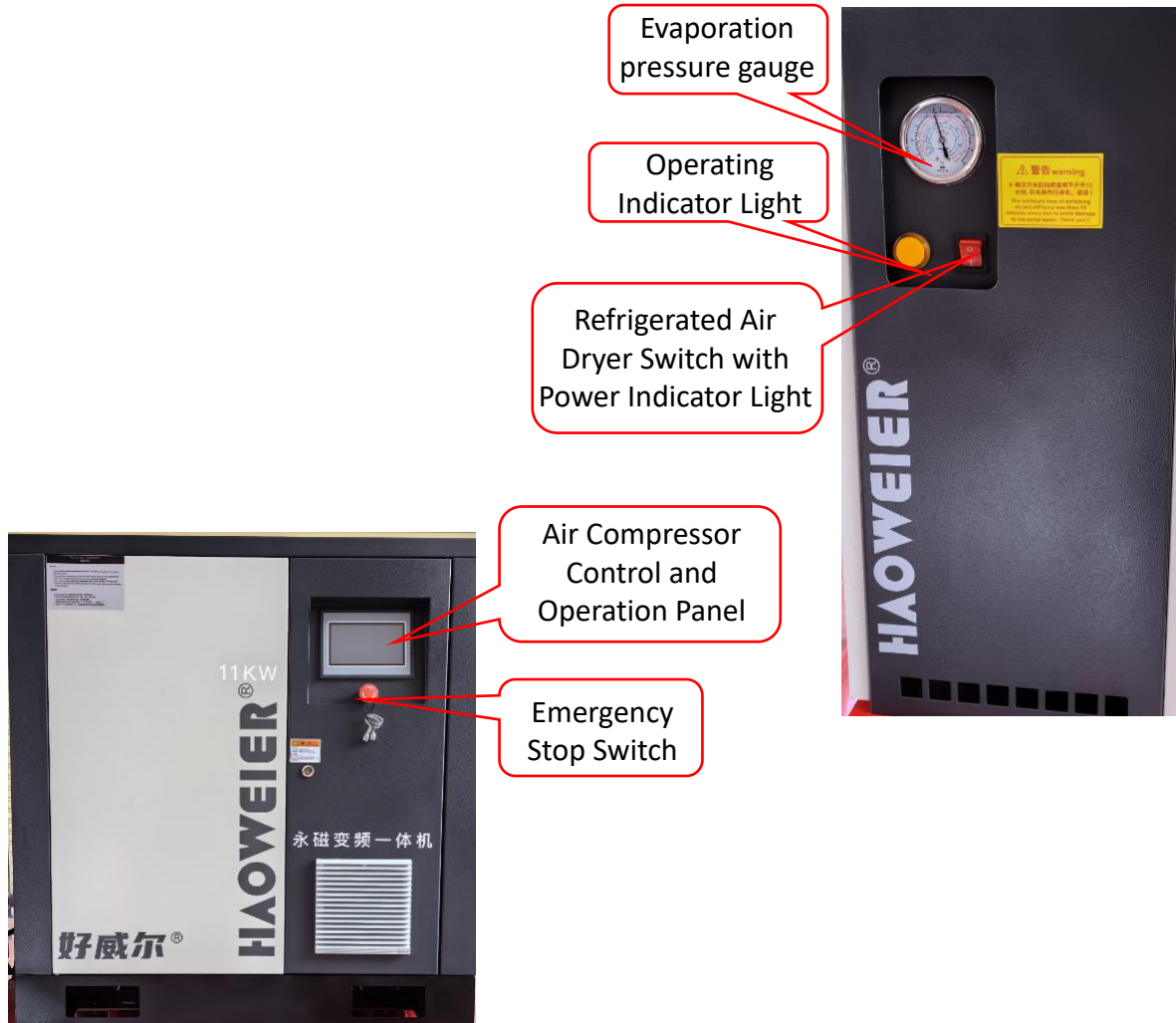
Structure Introduction Example (Back)

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Structural Overview Example (Control Section)

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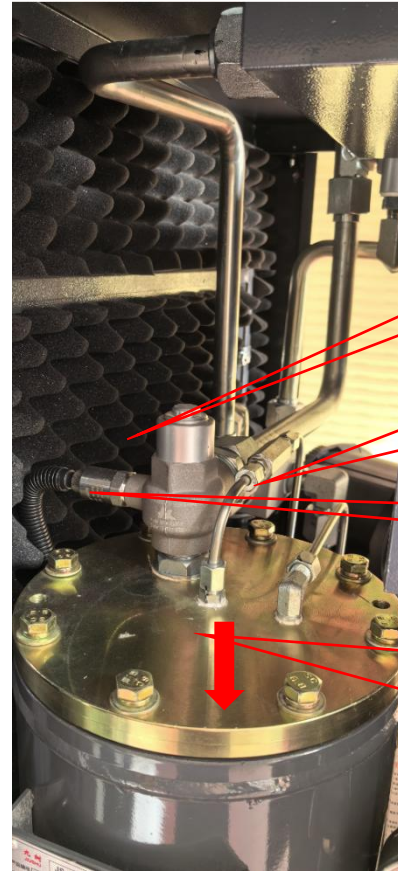
Structural Overview Example (Mechanical Section)

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Safety valve

Engine Oil
Filter Element

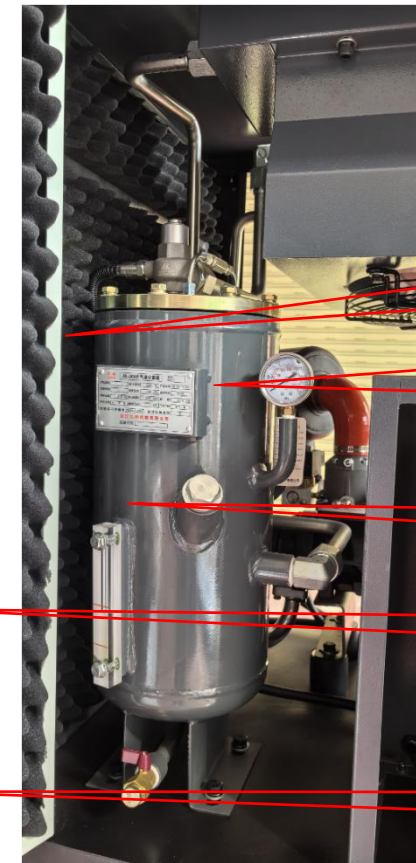


Minimum
Pressure Valve

Oil Return
Check Valve

Pressure
Transducer

Internally
Mounted Oil-Air
Separator
Element



Oil-Air Separator
Element (Internal)

Internal
pressure gauge

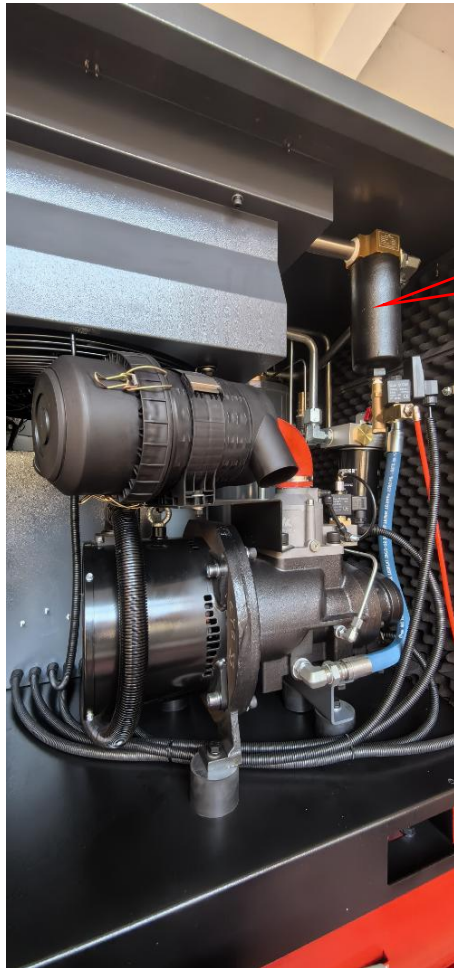
Filling Port

Oil Sight Glass

Unloading valve

Structural Introduction Example (Mechanical Part)

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Primary filter
22-37KW



Thermostatic
Valve Cartridge



Temperature
Sensor

22-37KW内部结构

Precautions for frequency converter

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WARNING: Failure to comply with the following may cause electric shock or other serious accidents!

Burns Hazard: Do not touch the radiator fins, transformers, or capacitors when they are hot, as this may cause burns.

Parameter Settings: Do not arbitrarily alter the VFD's factory preset parameters. Improper changes can damage the unit.

Electric Shock Hazard: The VFD carries high voltage. Never touch the VFD terminals while powered ON or before it has fully discharged after power-off, as this may lead to electric shock.

Power Isolation: Before any inspection or maintenance, the main power circuit **MUST** be disconnected. Ensure the charge indicator light is extinguished. Working on the VFD while its capacitors retain residual voltage is extremely dangerous.

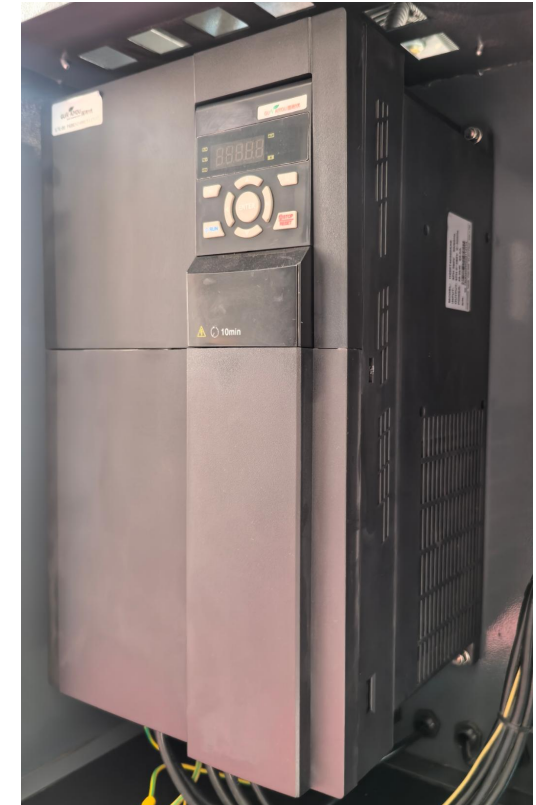
Qualified Personnel: Only qualified personnel are permitted to perform inspections, repairs, or part replacements. Remove all metal objects (e.g., watches, bracelets) beforehand and use only insulated tools to prevent electric shock.

Electrical Interference: The VFD is equipped with a DC reactor. If radio equipment or other electronic devices are installed nearby, fit a noise filter on the input power side to prevent interference.

ATTENTION!

Environmental Protection: The VFD contains sensitive components. It is crucial to ensure it remains dry and that dust is promptly removed. **The unit must be cleaned weekly to prevent overheating faults.**

Parameter Integrity: The internal parameters of the VFD are correctly calibrated at the factory. Unauthorized adjustment is strictly prohibited.



On-site installation

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The installation location and environmental conditions directly impact the later operational performance and cleanliness of the air compressor.

Environmental Requirements:

The air compressor should ideally be installed in a spacious, well-lit area (avoid direct sunlight if possible) to facilitate operation and maintenance.

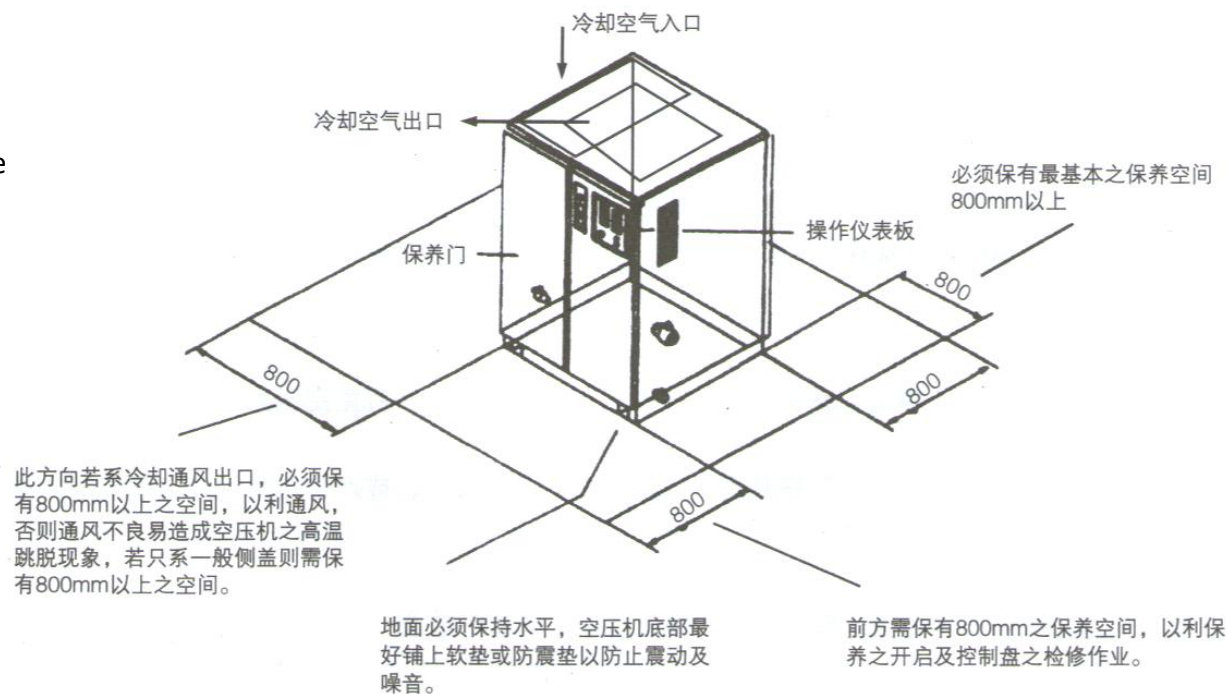
Air humidity should be as low as possible, with minimal dust, clean air, and good ventilation. Ambient temperature must be between 10° C and 40° C. Excessively high ambient temperature will lead to higher discharge air temperature from the compressor. Excessively low ambient temperature may cause the refrigerated air dryer to operate abnormally. Therefore, ambient temperature is crucial for the compressor's operating conditions.

Sufficient space must be reserved around the air compressor. A minimum distance of at least 70 centimeters is required between the compressor and any wall.

If the compressor is placed in an enclosed compressor room, a ventilation system must be installed (specific requirements to be determined by professionals based on the site conditions) to maintain the room temperature. The ventilation capacity of the room must exceed the airflow of the compressor's circulating fan or cooling fan, and the cold air intake area must be sufficient. Alternatively, a duct can be installed at the outlet of the compressor's exhaust fan to guide the hot air outside through the duct, thereby maintaining the room temperature.

Power Supply Requirements:

For 11-15KW air compressors, a 65A power switch is recommended; use a 6 mm² (3+2 core) cable.
For 22KW air compressors, a 100A power switch is recommended; use a 10 mm² (3+2 core) cable.
For 37KW air compressors, a 150A power switch is recommended; use a 16 mm² (3+2 core) cable.
For other requirements, please refer to the product manual!



Operation precautions

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Pre-Startup Precautions:

Before energizing, open the front and rear panels of the air compressor. Check for any foreign objects (tools, debris, etc.) inside the compressor cabinet.

With the power OFF, check for any loosening of internal components (e.g., oil filter element, oil pipe connection nuts) that may have occurred during transportation. Tighten them if necessary.

Connect the power supply according to the model's requirements. Measure to verify the voltage is correct, the neutral wire is secure, and the ground connection is proper.

Check that the air compressor's lubricating oil level is within the normal range.

Post-Energization Precautions:

Warning: After the air compressor is energized, non-qualified personnel must not touch any electrical components or Variable Frequency Drive (VFD) wiring!

After energization, confirm that all parameters on the display are normal and there are no alarm or warning indicators.

After startup, listen for any unusual noises inside the compressor and observe whether all parameters on the display are reading normally.

The unit is ready for normal operation after it completes one full cycle of pressure rise and fall without any abnormalities.

Routine Maintenance Items:

Ensure the area around the air compressor remains well-ventilated. Control the ambient temperature to not exceed 40° C nor fall below 5° C. Minimize dust ingress as much as possible.

Clean internal and external dust from the air compressor weekly. Pay particular attention to the internal cooler/radiator, the intake air filter for the permanent magnet motor, and the intake grilles.

Manually drain condensate from the air receiver 2-3 times daily.

Check the operation of the electronic drain valve weekly to ensure it functions correctly.

For matters not covered here, please refer to the product manual.



New machine startup:

Note: Initial Startup of New Machine

Due to prolonged static storage or long-distance transportation, the internal piping of the machine may become contaminated due to condensation, pipe nuts may occasionally loosen, and electrical cabinet wiring connections might also be found loose. Therefore, the following checks are essential before startup:

Inspect all pipeline connections. Tighten all connection nuts, pipe clamps, and other fasteners.

Check for any loose wiring inside the electrical cabinet and within the machine itself.

After starting, carefully observe the machine for any internal air or oil leaks.

When starting the compressor, the refrigerated air dryer must be started simultaneously. Observe whether the dryer's fan is running, its front indicator light is on, and its temperature display is functioning normally. **Once pressure reaches above 5 bar, start the desiccant air dryer** (press the ON/OFF key, observe if the two pressure gauges switch normally and if the bottom muffler/silencer depressurizes correctly).

During the initial startup, after the pressure reaches the upper limit, open the ball valve at the compressor's final discharge outlet by one-third. Allow continuous air blowing (purging) for **over 2 hours** before connecting the pipeline to the laser cutting equipment (It is recommended that after the cutting head position is connected to the air, it should be blown empty for about 20 minutes simultaneously to clean the bed pipeline and valve components) .

Main operating interface

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Home Screen (Startup Screen)

This interface is the home screen displayed upon startup. It provides a direct overview of the air compressor's current status, including:

Pressure, Temperature, Frequency, Current, Voltage, Operating Status, Total Running Hours

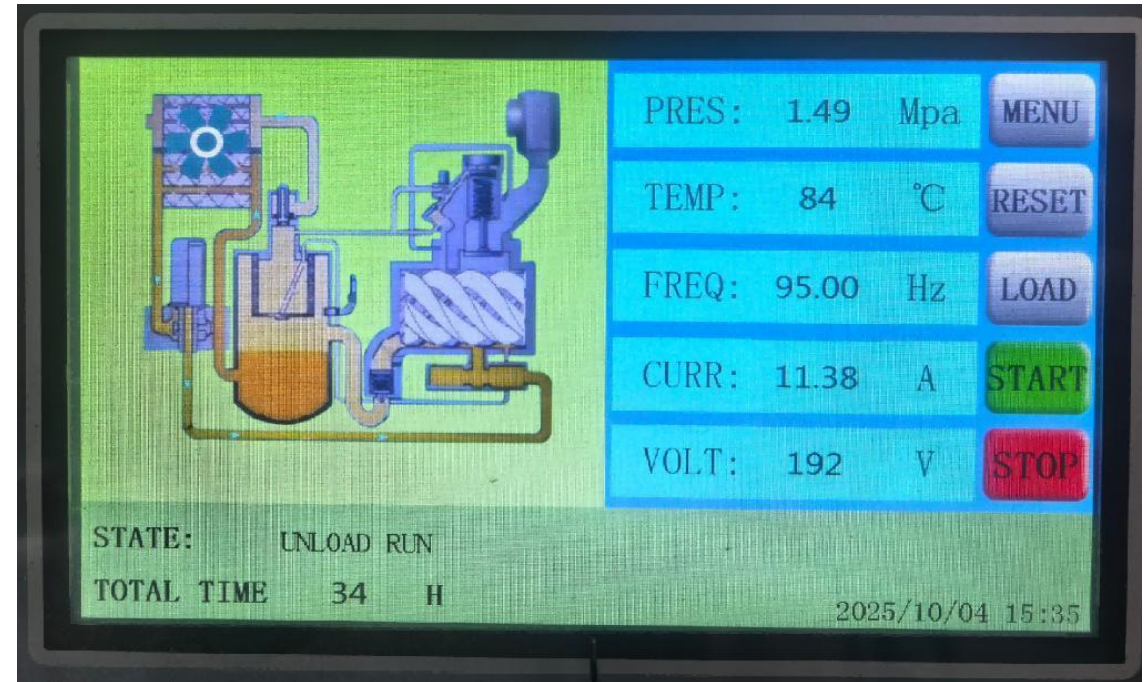
Menu: Tap to enter the secondary interface.

Reset: After an alarm is cleared or a fault is resolved, press and hold for 3 seconds to reset parameters.

Load: (Manual Mode) Tap once to initiate loading; tap again to unload.

Start: Press and hold for 3 seconds to start the air compressor.

Stop: Tap to shut down the air compressor.



Controller parameter list

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Secondary Parameter Interface

Operating Parameters: Access to view the current operational data of the air compressor.

User Parameters: Access to modify parameters within the user's authority level.

Factory Parameters: These are preset parameters configured by the manufacturer. Modification is not recommended. Please consult your supplier if you have any questions!

Consumable Parameters: Access to monitor and maintain the service life of consumable components.

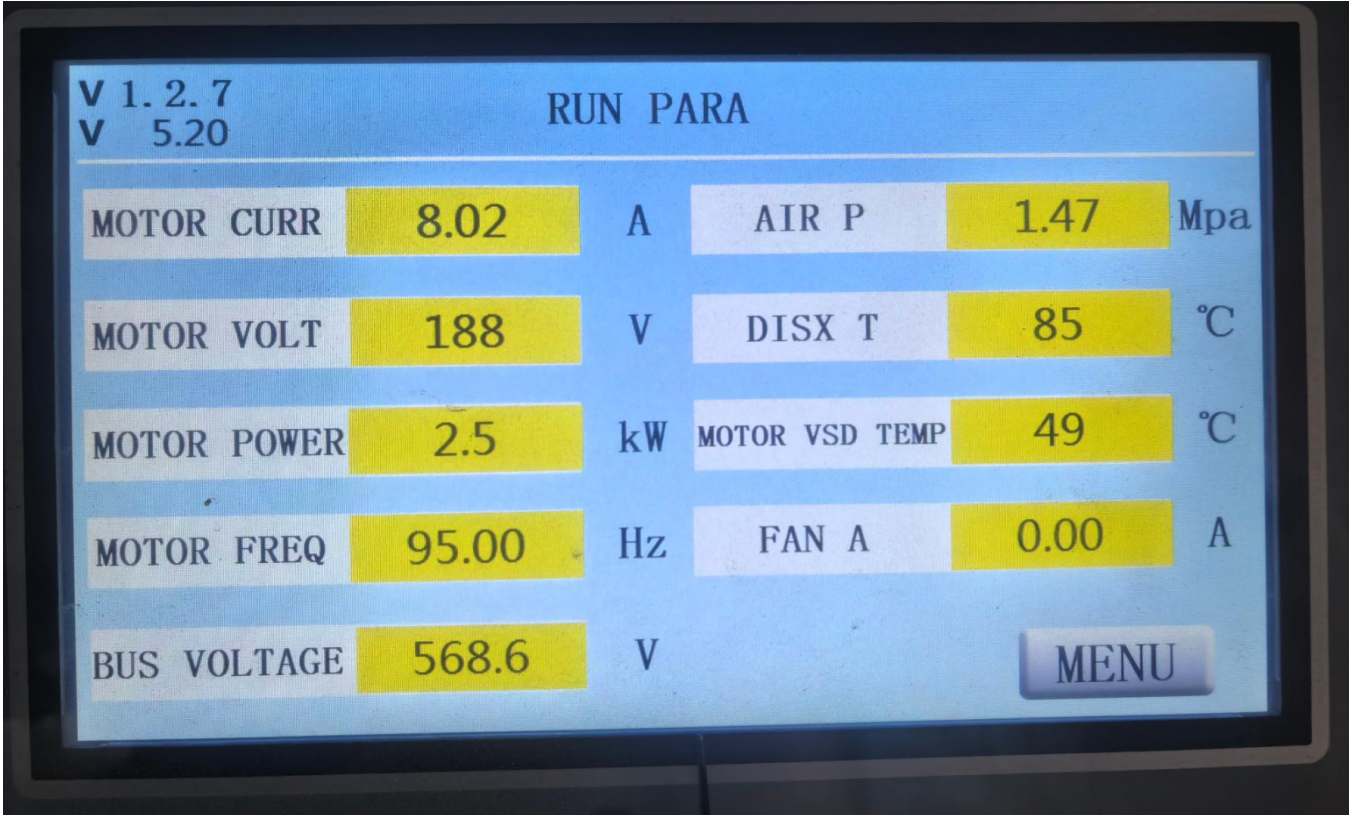
Fault Log: Access to view the historical record of recent faults encountered by the air compressor.

Note: Do not arbitrarily access any interfaces not listed here!



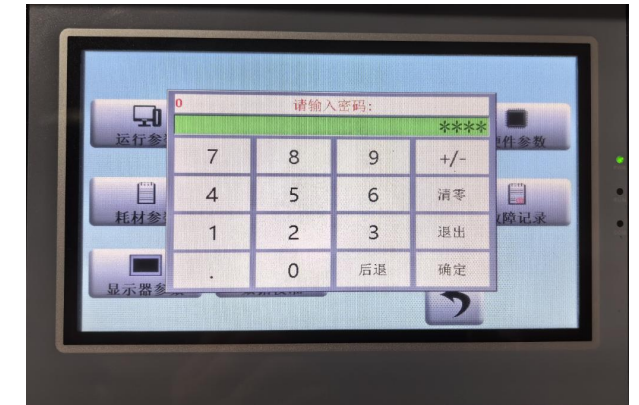
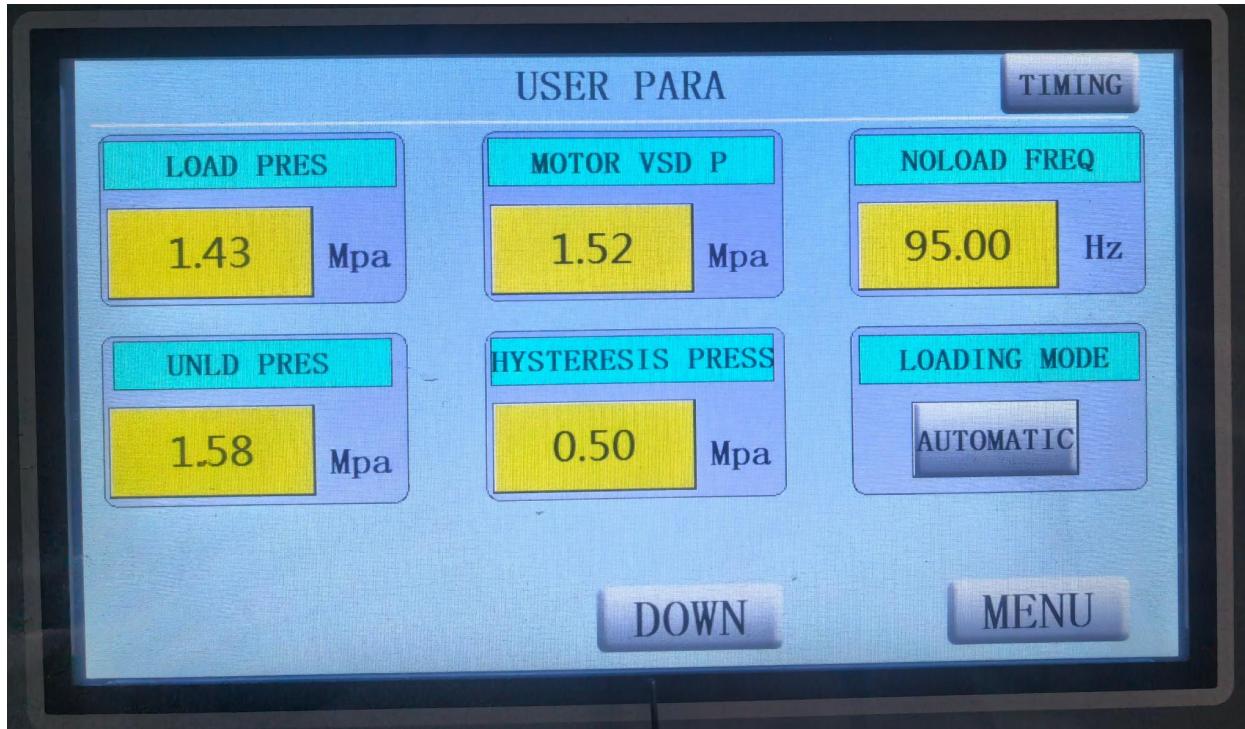
Secondary Interface - Main Unit
Operating Information

This screen provides a direct overview of the air compressor's key operating parameters.
Press the 'Menu' key to return to the home screen.



User Parameter Interface 1

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Secondary Interface - User Parameters

This interface allows you to modify various operating parameters of the air compressor.

Load Pressure Setpoint: Sets the minimum required pressure for the air compressor to start supplying air.

Unload Pressure Setpoint: Sets the maximum required pressure for the air compressor to stop supplying air.

Constant Speed Pressure: Sets the constant pressure maintained by the air compressor when operating at fixed speed.

No-Load Frequency: Sets the compressor's operating frequency under no-load conditions. (Factory preset. Unauthorized modification is strictly prohibited.)

Press the 'Menu' key to return to the home screen.

User Parameter Interface 2

Secondary Interface - User Parameters

This interface allows you to modify various operating parameters of the air compressor.

No-Load Delay Time: Sets the delay time before the compressor shuts down after unloading. (Recommended: 90-360 seconds)

Restart Delay Time: Sets the delay time required before the compressor can be restarted after shutdown. (Recommended: 5-10 seconds)

Shutdown Delay Time: Sets the delay time before the compressor stops after a shutdown command is issued. (Recommended: 10-30 seconds)

Language: Toggles the display language between English and Chinese with each press.

Press the 'Menu' key to return to the home screen.



Manufacturer parameter interface 1

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Secondary Interface - Factory Parameters

This interface allows modification of the air compressor's factory parameters. (These parameters are preset at the factory. Unauthorized changes are not recommended. Please consult your supplier for the access password if required.)

Exhaust Temperature Warning: The compressor will trigger a warning alert when it reaches this temperature. (Limit: 105° C)

Exhaust Temperature Shutdown: The compressor will perform an emergency shutdown when it reaches this temperature. (Limit: 110° C)

Supply Pressure Shutdown: The compressor will perform an emergency shutdown when it reaches this pressure. (Limit: 1.63 MPa)

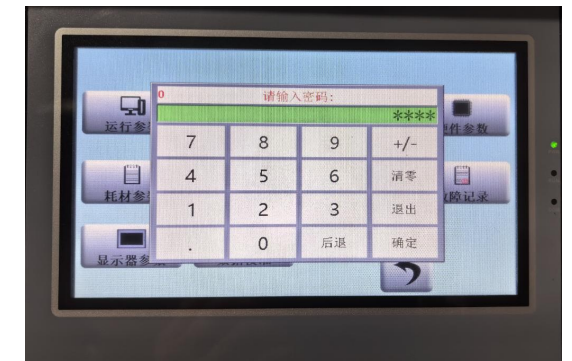
Prolonged Warning Shutdown: The compressor will perform an emergency shutdown if a warning condition persists for this duration. (Factory default: No restriction)

Power Factor: Refer to the motor nameplate for this value.

Low-Temperature Protection: The compressor will perform an emergency shutdown when the ambient temperature reaches this level. (Limit: -10° C to -30° C)

Unload High-Limit Pressure: The maximum allowable system pressure. Upon reaching this value, the compressor will forcibly unload. (Limit: 1.6 MPa)

Press the 'Menu' key to return to the home screen.



Manufacturer parameter interface 2

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Secondary Interface - Factory Parameters

This interface allows modification of the air compressor's factory parameters. (These parameters are preset at the factory. Unauthorized changes are not recommended. Please consult your supplier for the access password if required.)

Total Run Time Optimization: Records the total operational hours of the air compressor.

Loaded Run Time: Records the compressor's operational hours under load.

Low-Temperature Run Time: Sets the allowable operating duration within low-temperature conditions.

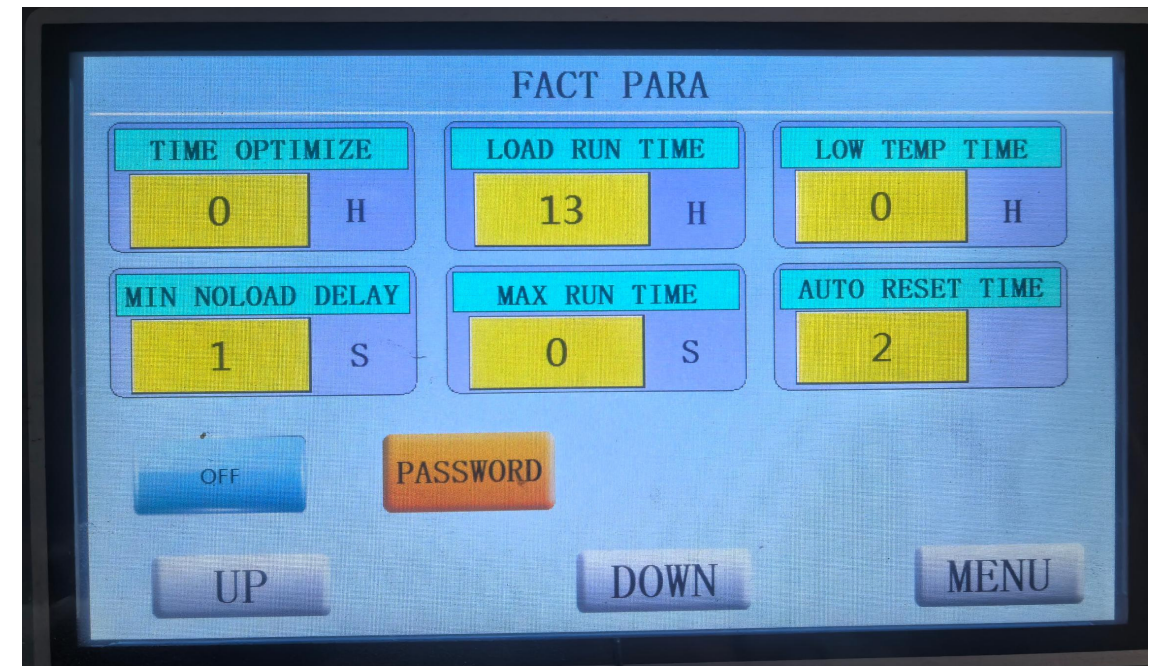
Minimum No-Load Time: Sets the minimum delay time for no-load operation.

Maximum Run Time: Sets the maximum permitted operational duration for the compressor.

Auto Reset Count: Sets the number of automatic reset attempts allowed.

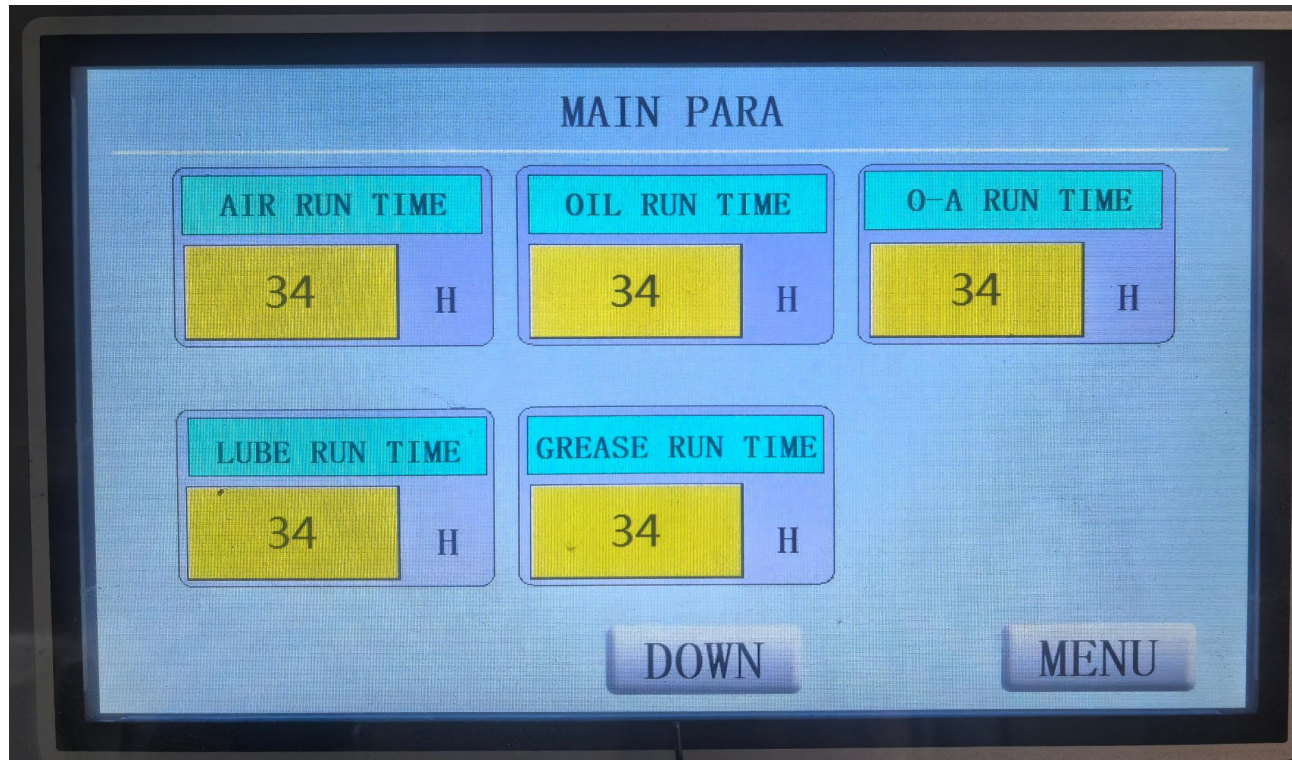
Debug Screen: For factory testing purposes only. Strictly prohibited for unauthorized or non-professional use.

Press the 'Menu' key to return to the home screen.



Consumables Parameters Interface 1

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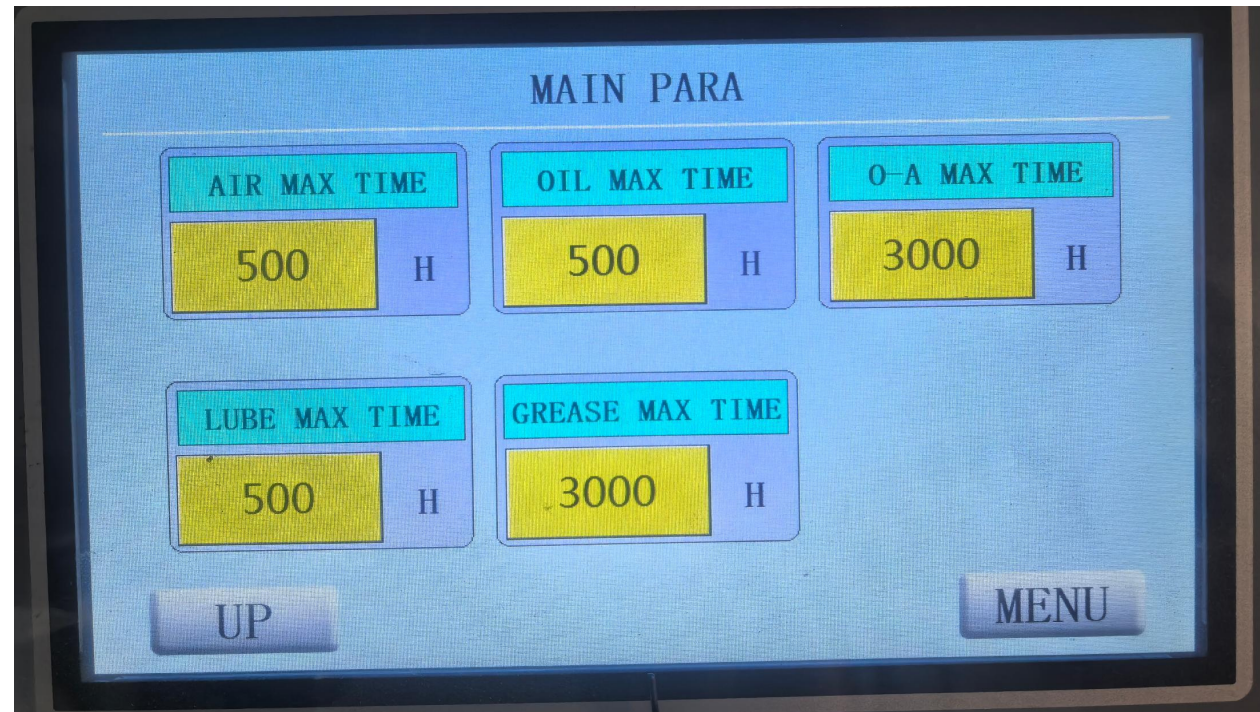


Secondary Interface - Consumable Parameters

This interface displays the current usage time of consumables. Authorized personnel can view, modify, and reset the service timers for the air compressor's consumable components. Password access is required.

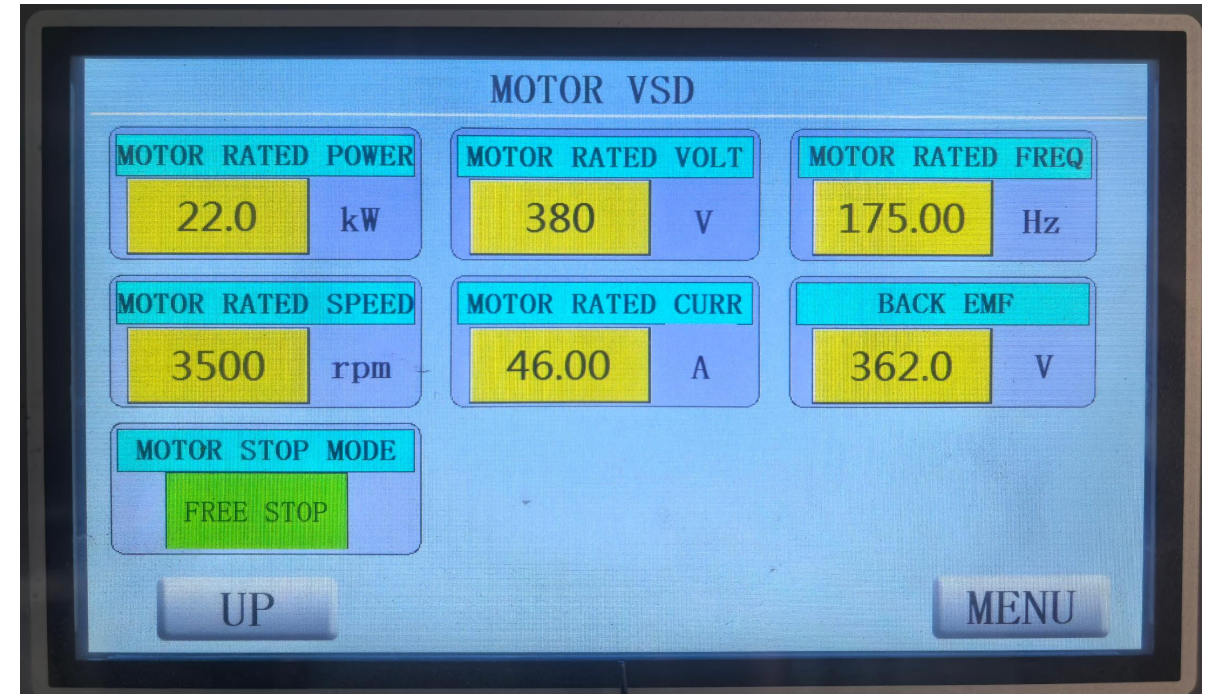
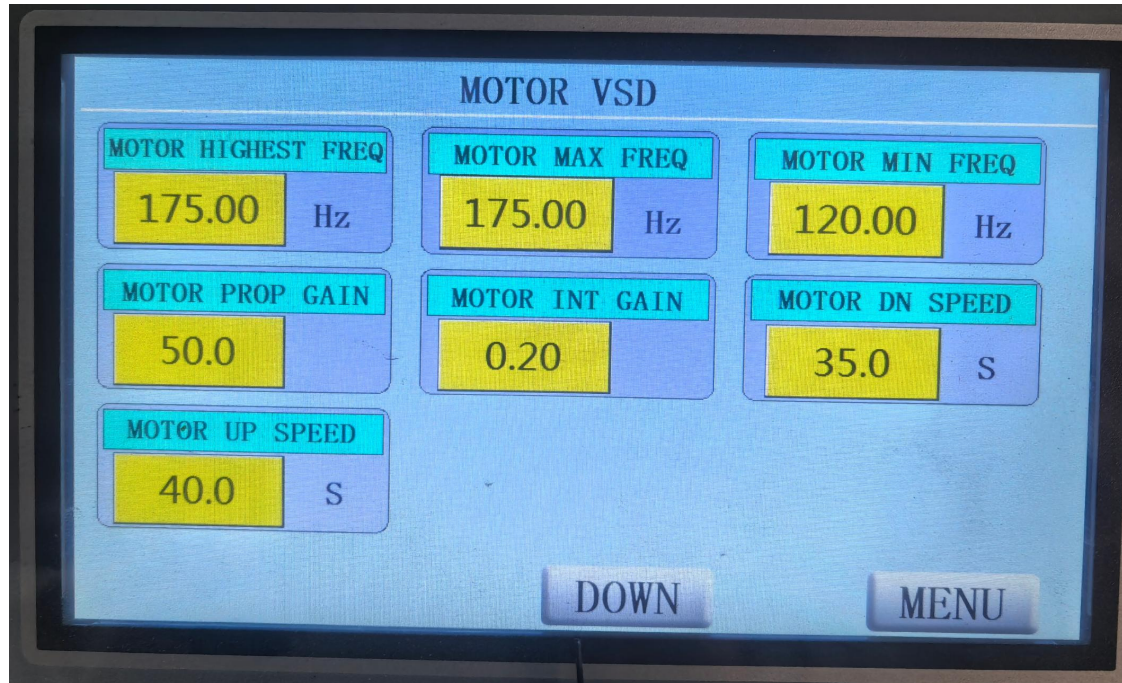
Secondary Interface - Consumable Parameters

This interface displays the preset service timers for consumables. All usage durations for the air compressor's consumable components can be set here. Press the 'Menu' key to return to the menu interface.



•Primary Frequency Conversion Parameter Interface

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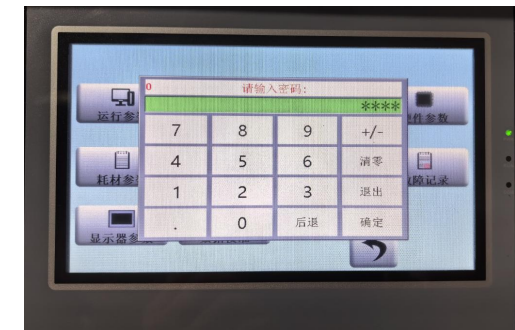


Secondary Interface - Main VFD Parameters

Warning: These parameters are factory preset. Modification is strictly prohibited without manufacturer authorization.

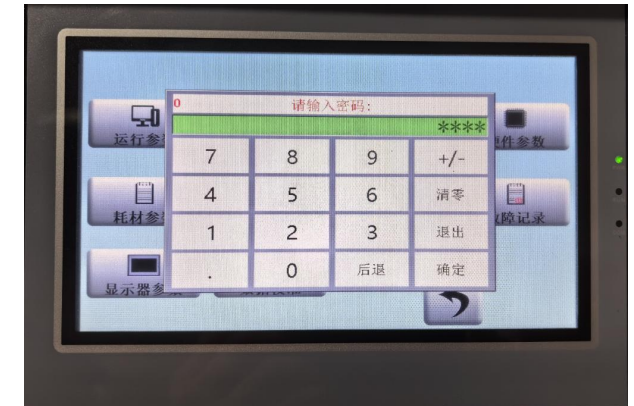
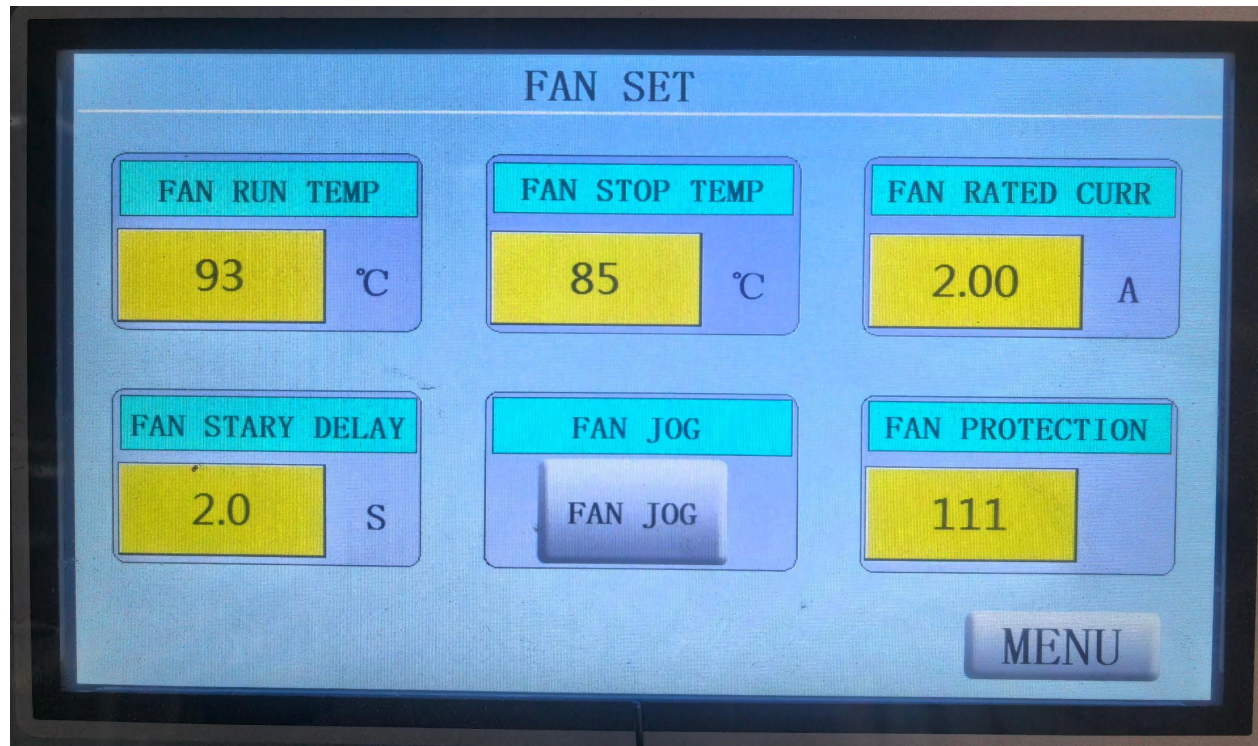
This interface allows configuration of the air compressor's power, voltage, speed, and the VFD's upper/lower frequency limits.

Press the 'Menu' key to return to the menu interface.



Fan setting parameter interface

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Secondary Interface - Fan Settings

Warning: These parameters are factory preset. Modification is strictly prohibited without manufacturer authorization. This interface allows configuration of the air compressor's cooling fan start/stop temperature and current settings, as well as testing fan functionality.

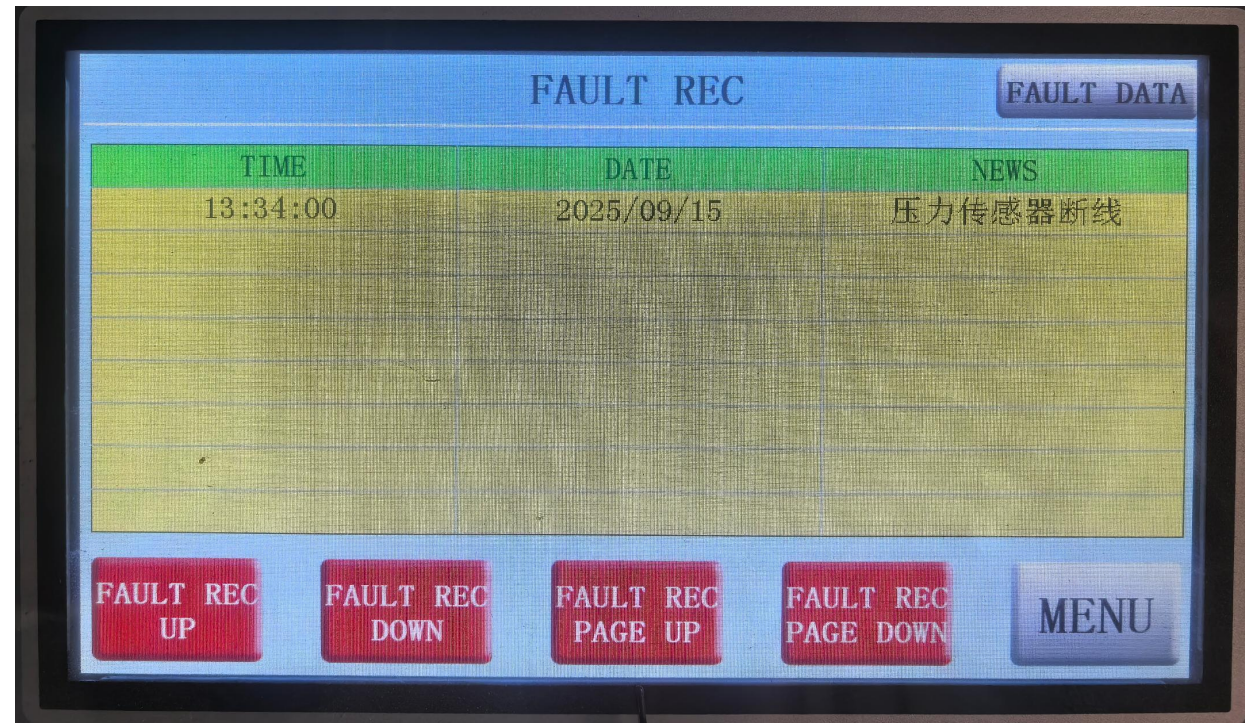
Fan Jog: Used during initial startup or after power phase changes. This function tests the correct rotational direction of the fan assembly.

Press the 'Menu' key to return to the menu interface.

Secondary Interface - Fault History

This interface displays recent historical faults of the air compressor, enabling targeted problem diagnosis.

Fault Data: Tap to view detailed alarm data recorded at the time of the warning. Press the 'Menu' key to return to the menu interface.

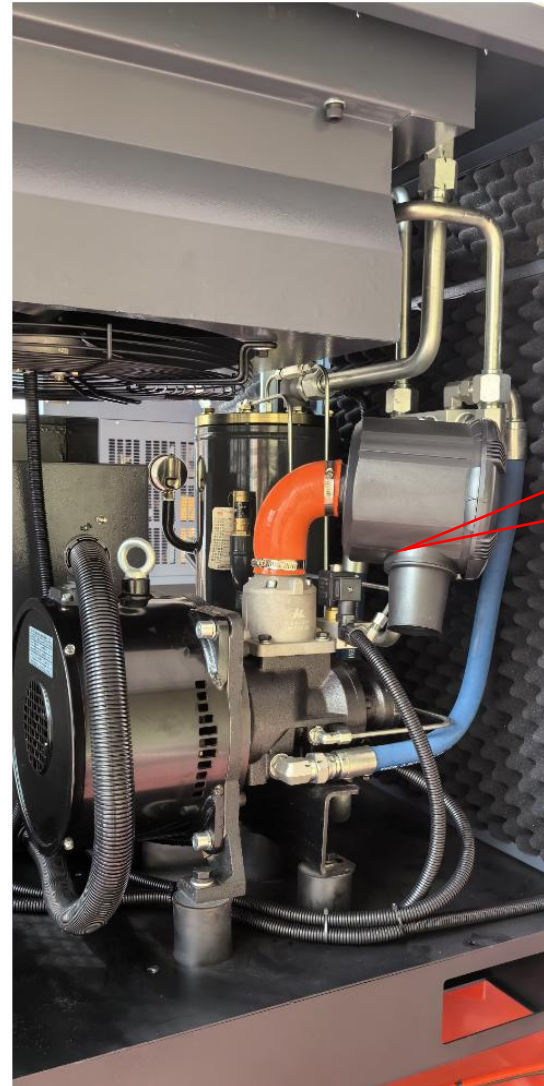


Maintenance

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Oil Filter Element:
Initial Replacement:
500 operating hours.
Subsequent Replacement:
Every 3,000 operating hours or 6 months, whichever comes first.



Air Filter Element:
Initial Replacement: 500 operating hours.
Subsequent Replacement: Every 3,000 operating hours or 6 months, whichever comes first.

Maintenance

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Permanent Magnet VFD Model - Internal Oil Separator Replacement

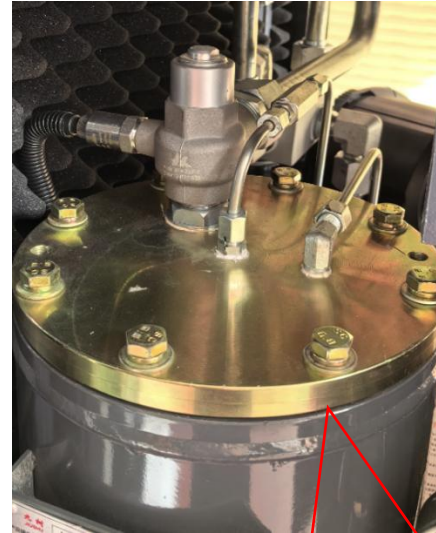
Procedure:

Drain oil completely before disassembly.
Remove external pipes from the separator cover.
Loosen all perimeter screws and remove the cover.
Extract the old oil separator element.
Inspect the internal tank for contamination/rust and clean if needed.

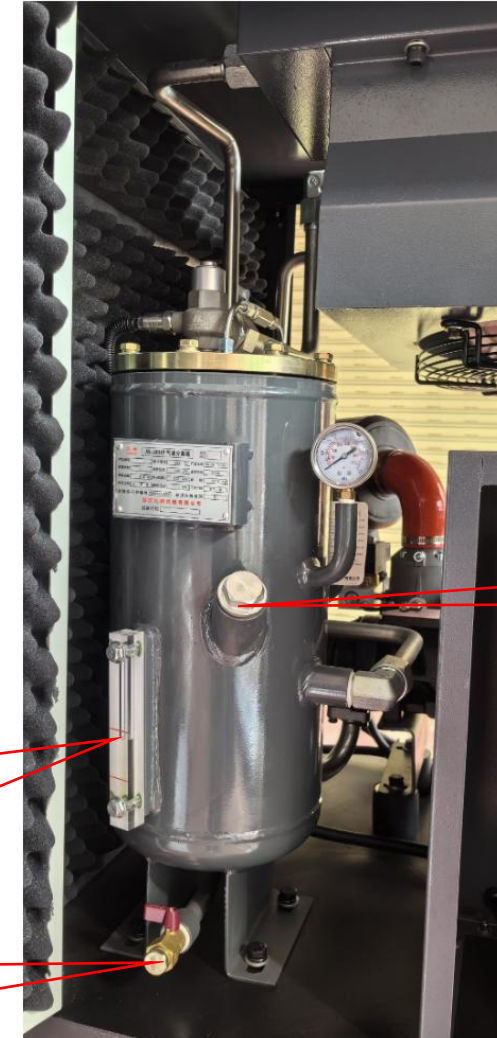
Install new element: Preserve the staple on the gasket.
Apply a thin layer of lubricant to the gasket.
Reinstall cover, aligning screw holes. Tighten screws diagonally and evenly.
Reconnect all pipes and components.

Oil Change:

Refer to diagram for oil fill and drain port locations.
Use only specified lubricant type.



The internal oil separator requires removal of the separator cover for replacement.



During maintenance, fill lubricant only to the upper red line. Do not exceed this level to prevent excessive oil content in the air.

Add lubricant at this location.

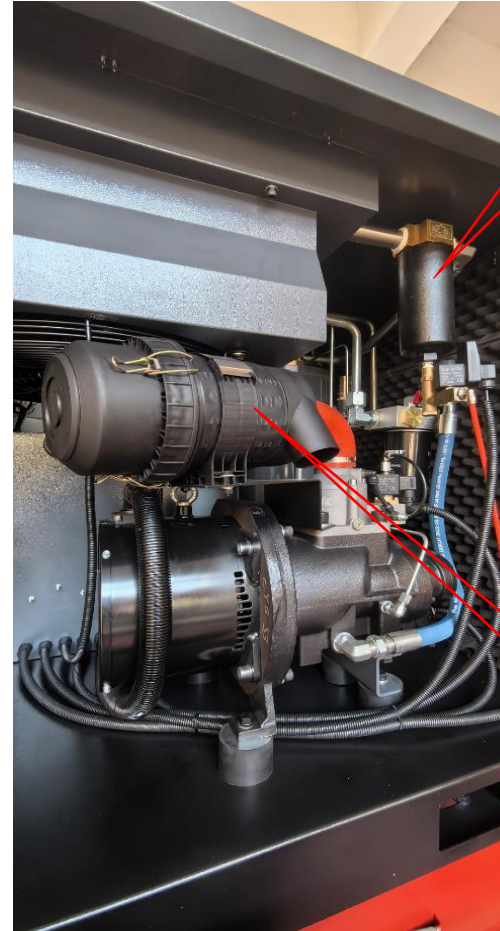
Drain Valve: Waste oil is discharged from this location.

Maintenance

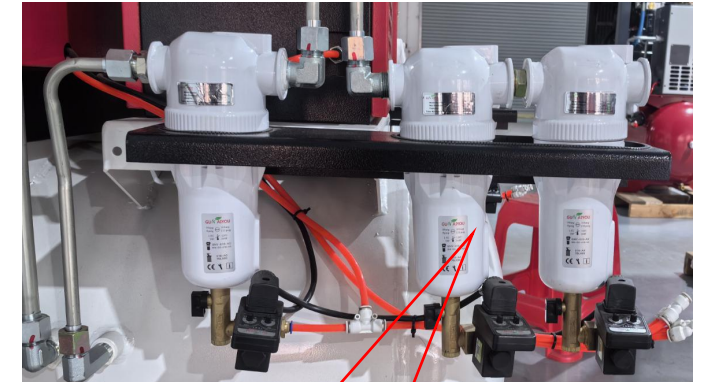
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Filter Element & Primary Filter Replacement:

Warning: Ensure pressure is fully released to 0 and the electronic drain valve is powered OFF before disassembly. Remove the electronic drain valve at the housing base before replacing the primary filter. Replace with new filter element after removing the housing, ensuring the seal is properly seated. When replacing subsequent multi-stage filters, maintain correct sequence and avoid mixing different grades. Verify all sealing rings remain correctly positioned during installation. Reassemble housing and electronic drain valve after element replacement.



Primary filter



Multi-Stage Filter Location

Air filter

Maintenance

Important Notes:

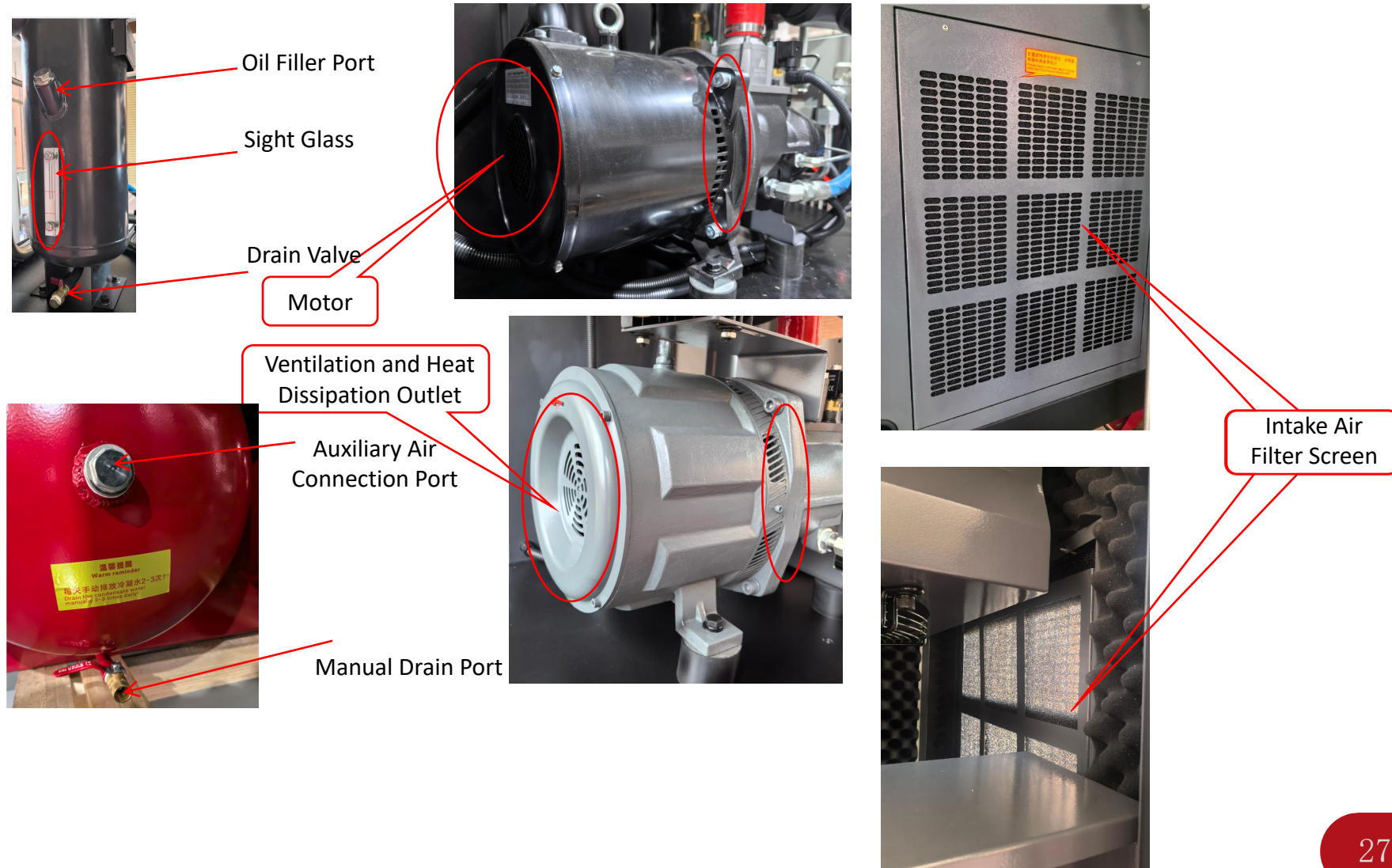
The intake air filter screen must be cleaned weekly by blowing compressed air from the inside out.

The permanent magnet motor must be cleaned weekly by blowing compressed air inward through the side intake vents to maintain proper internal ventilation and heat dissipation.

The compressor is filled with qualified lubricant at the factory. During maintenance, only use the manufacturer-specified lubricant. Mixing different brands of lubricants is strictly prohibited. When the unit is idle, the oil level should be maintained between the upper and lower red lines on the sight glass.

Open the manual drain valve 2-3 times daily to discharge condensate, ensuring no significant amount of condensate accumulates in the air receiver.

Note: The translation maintains the original technical meaning while adapting to standard English technical documentation style. Key safety instructions are clearly emphasized.



Abnormal pressure handling

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Note: Before disassembly, make sure to release the air pressure!! Operate only after power off.
After maintenance is completed, restore power and restart.

Fault point: The controller shows that the pressure is higher or lower than the actual pressure. The controller indicates that the pressure has not reached the unloading pressure, but the actual pressure has already reached the mechanical protection pressure value of the safety valve and triggered its action.

Cause of the fault: This issue mainly occurs when the environmental humidity is high, and the condensate water generated after the hot machine stops cannot be discharged in time, or when the usage frequency is low and the machine has been idle for a long time, the sensitive probe inside the pressure sensor (Figure 2) is interfered by the condensate water, causing incorrect data reading.

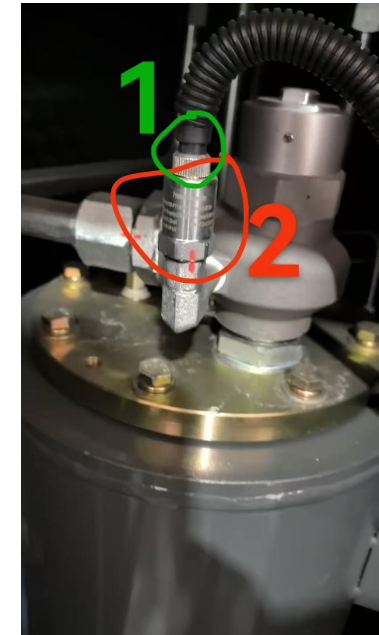
Solution: The pressure sensor needs to be removed (Figure 4), and the condensate water inside the detection hole at the bottom of the pressure sensor should be cleaned with an air gun. Then, it should be reinstalled (Figure 3).



图二



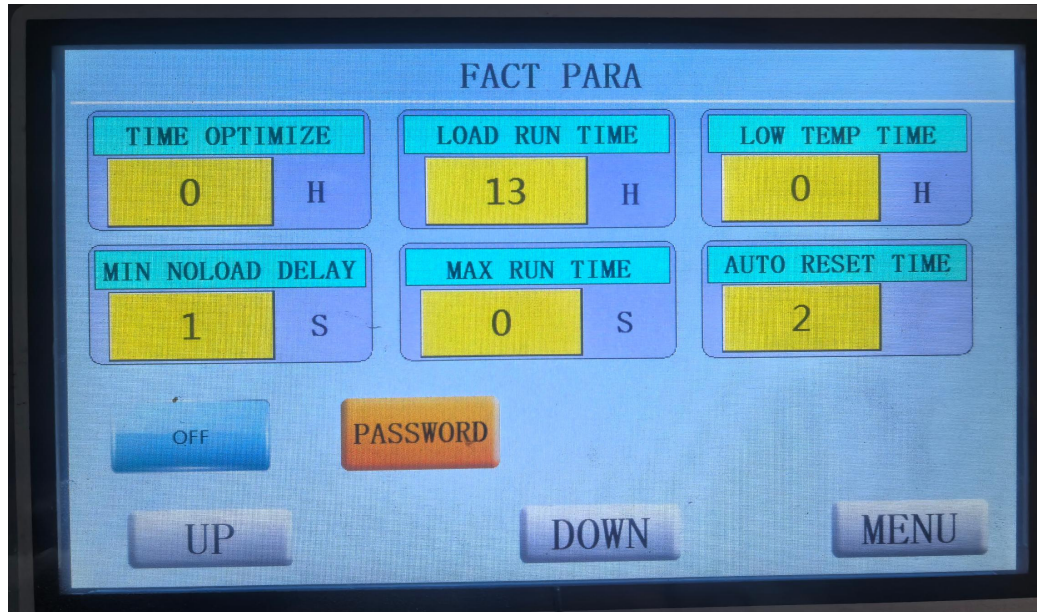
图三



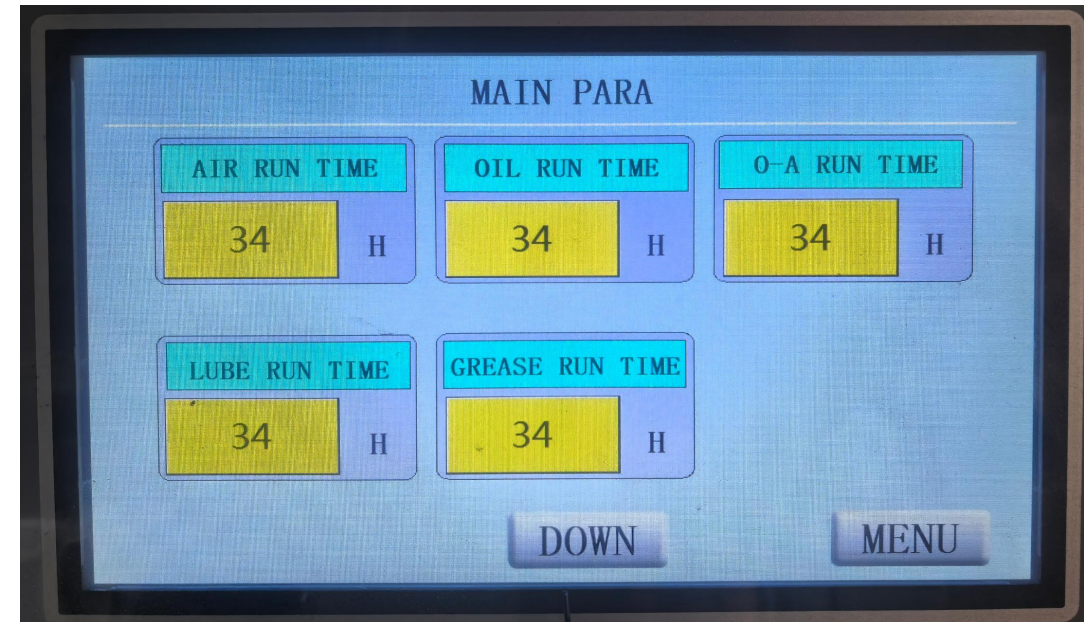
图四

Consumable Maintenance Parameter Reset

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Reset all consumable service timers to zero in the above diagram.



Set all consumable preset times to 3000 hours in the diagram above.

Consumables List for Common Permanent Magnet Variable Frequency Models



11-15KW

Consumables List

HAOWEIER 11-15kw integrated compressor 500-3000H Spare parts kit list (HB11PM-16-300/HB15PM-16-300)

NO.	Material code	Material specification	Name	Unit	Number	Replace time	Note
1	68082909	11-15KW	Air filter	pcs	1	3000hours or 6 months whichever comes first	Replace after 500 hours of first maintenance
2	68083506	11-15KW	Oil filter	pcs	1		
3	11020004	10L	Oil for screw machine	pcs	1		
4	68083501	11-15KW	Oil-gas separator	pcs	1		
5	10020401	GNY-015-AO	Filter element	pcs	1		Upgrade 018 in May 2023
6	10020402	GNY-015-AA	Filter element	pcs	1		
7	10020403	GNY-015-AX	Filter element	pcs	1		
8	10020421	GNY-018-AO	Filter element	pcs	1		Used after June 2023
9	10020422	GNY-018-AA	Filter element	pcs	1		
10	10020423	GNY-018-AX	Filter element	pcs	1		
11	13020370	11-15KW	Primary filter cartridge	pcs	1		Pay attention to the specifications and models when placing an order

Note:

1. The first 500 hour maintenance requires replacement of the air filter element, oil filter element, and lubricating oil; Replace consumables every 3000 hours or 6 months in the later stage (whichever comes first).
2. Before May 2023, all 015 series precision filters will be used. Please confirm the label model on the lower housing of the filter when purchasing.
3. Starting from May 2023, the 015 series filters will be upgraded to the 018 series filters. When purchasing, please confirm the label model on the lower housing of the filter.
4. The filter cartridges used in primary filters are divided into new and old models. When purchasing, carefully confirm the model identification on the filter housing to avoid selecting the wrong filter cartridge model!

22KW

Consumables List

HAOWEIER 22kw integrated compressor 500-3000H Spare parts kit list (HB22PM-16-300)

NO.	Material code	Material specification	Name	Unit	Number	Replace time	Note
1	68082910	22-37KW	Air filter	pcs	1	3000hours or 6 months whichever comes first	Replace after 500 hours of first maintenance
2	68083504	22-37KW	Oil filter	pcs	1		
3	11020004	10L	Oil for screw machine	bucket	1		
4	68083502	22KW (built-in)	Oil-gas separator	pcs	1		Upgrade 035 in May 2023
5	10020431	GNV-034-AO	Filter element	pcs	1		
6	10020432	GNV-034-AA	Filter element	pcs	1		
7	10020433	GNV-034-AX	Filter element	pcs	1		
10	10020441	GNV-035-AO	Filter element	pcs	1		Used after June 2023
11	10020442	GNV-035-AA	Filter element	pcs	1		
12	10020443	GNV-035-AX	Filter element	pcs	1		
15	13020371	22KW	Primary filter element	pcs	1		

Note:

1. The first 500 hour maintenance requires replacing the air filter element, oil filter element, and lubricating oil (which can be flexibly controlled according to the oil quality); Replace consumables every 3000 hours or 6 months in the later stage (whichever comes first).
2. Before May 2023, all 034 series precision filters will be used. Please confirm the label model on the lower housing of the filter when purchasing.
3. Starting from May 2023, the 034 series filters will be upgraded to the 035 series filters. When purchasing, please confirm the label model on the lower housing of the filter.
4. The filter cartridges used in primary filters are divided into new and old models. When purchasing, carefully confirm the model identification on the filter housing to avoid selecting the wrong filter cartridge model!

Hoverr 22KW Consumables for Integrated Twin-Tower Desiccant Air Dryer Unit (TGJ-30VSD-16-500)

HAOWEIER 22kw Dual tank integrated compressor 500-3000H Spare parts kit list (TGJ-30VSD-16-500)

NO.	Material code	Material specification	Name	Unit	Number		Replace time	Note
1	68082910	22-37KW	Air filter	pcs	1		3000 hours or 6 months whichever comes first	Replace after 500 hours of first maintenance
2	68083504	22-37KW	Oil filter	pcs	1			
3	11020004	10L	Oil for screw machine	bucket	1			
4	68083502	22KW (built-in)	oil-gas separator	pcs	1			Replace after 3000 hours or 6 months
5	10020441	GNY-035-AO	Filter element	pcs	1			
6	10020442	GNY-035-AA	Filter element	pcs	1			
7	10020443	GNY-035-AX	Filter element	pcs	2			
8	10020445	GNY-035-AAR	Filter element	pcs	1			
9	10020444	GNY-035-ACS	Filter element	pcs	1			
10	13020371	22-37KW	Primary filter element	pcs	1			
16	12053208	GNY-16KG	Electronic drain valve	pcs	1		spare parts	

Note:

The first 500 hour maintenance requires replacement of the air filter element, oil filter element, and lubricating oil (which can be flexibly controlled according to the oil quality); Replace consumables every 3000 hours or 6 months in the later stage (whichever comes first).

Hoverr 37KW Consumables for Integrated Twin-Tower Desiccant Air Dryer Unit (TGJ-50VSD-16-500)

HAOWEIER 37kw Dual tank integrated compressor 500-3000H Spare parts kit list (TGJ-50VSD-16-500)

NO.	Material code	Material specification	Name	Unit	Number	Replace time	Note
1	68082910	22-37KW	Air filter	pcs	1	3000 hours or 6 months whichever comes first	Replace after 500 hours of first maintenance
2	68083504	22-37KW	Oil filter	pcs	1		
3	11020004	10L	Oil for screw machine	bucket	2		
4	13020374	37KW (bulit-in)	Oil-gas separator	pcs	1		Replace after 3000 hours or 6 months
5	10020441	GNY-035-AO	Filter element	pcs	1		
6	10020442	GNY-035-AA	Filter element	pcs	1		
7	10020443	GNY-035-AX	Filter element	pcs	2		
8	10020445	GNY-035-AAR	Filter element	pcs	1		
9	10020444	GNY-035-ACS	Filter element	pcs	1		
10	13020371	22-37KW	Primary filter element	pcs	1		
16	12053208	GNY-16KG	Electronic drain valve	pcs	1	Spare parts	

Note:

The first 500 hour maintenance requires replacement of the air filter element, oil filter element, and lubricating oil (which can be flexibly controlled according to the oil quality); Replace consumables every 3000 hours or 6 months in the later stage (whichever comes first).

37KW Consumables List

HAOWEIER 37kw Integrated compressor 500-3000H Spare parts kit list (TGX-50VSD-16-2600)							
NO.	Material code	Material specification	Name	Unit	Number	Replace time	Note
1	68082910	22-37KW	Air filter	pcs	1	3000 hours or 6 months whichever comes first	Replace after 500 hours of first maintenance
2	68083504	22-37KW	Oil filter	pcs	1		
3	11020004	10L	Oil for screw machine	bucket	2		
4	13020374	37KW (built-in)	oil-gas separator	pcs	1		Replace after 3000 hours or 6 months
5	10020446	GNY-065-AO	Filter element	pcs	1		
6	10020447	GNY-065-AA	Filter element	pcs	1		
7	10020448	GNY-065-AX	Filter element	pcs	2		
8	10020449	GNY-065-AAR	Filter element	pcs	1		
9	10020450	GNY-065-ACS	Filter element	pcs	1		
10	13020371	22-37KW	Primary filter element	pcs	1		Optional: high-precision gas source configuration
11	09080404	GNY-CM5.0/A	Filter element	set	1		
12	09080405	GNY-CM5.0/T	Filter element	set	1		
13	09080406	GNY-CM5.0/H	High precision oil removal filter	set	1		Optional: special for cutting head protection
14	09120110	GNY-026-AX	High precision filter	set	1		
15	10080111	GNY-026AX	High precision filter element	pcs	1		
16	12053208	GNY-16KG	Electronic drain valve	pcs	1		
Note: 1. The first 500 hour maintenance requires replacing the air filter element, oil filter element, and lubricating oil (which can be flexibly controlled according to the oil quality); Replace consumables every 3000 hours or 6 months in the later stage (whichever comes first).							

Permanent Magnet Variable Frequency Series - Product Overview



1000-6000W Laser Cutting Standard Integrated Unit

NO	Technical Parameters		
1	Model	HB11PM-16-300	HB15PM-16-300
2	Air Delivery Capacity	1.0m³/min	1.35m³/min
3	Working pressure	1.6MPa	
4	Transmission Type	Direct-Drive Variable Frequency	
5	Cooling method	Air cooling	
6	Power	11KW	15KW
7	Duty Cycle Rating	Continuous Operation	
8	Power supply	380V/50HZ/3PH	
9	Air Connection Port Size	G½"-φ10/φ12	
10	Rotation speed	2750	
11	Air tank Capacity	350L	
11	Weight	420KG	440KG
12	Length × Width × Height	1550*780*1680	
Compatible Laser Power		1000-3000W	4000-6000W



Model Advantages:

Integrated Design: Incorporates permanent magnet variable frequency screw compressor, air treatment equipment, and air receiver into a single compact unit.

Space-Saving: Optimized footprint specifically facilitates installation at laser cutting facilities.

Superior Core Components: Utilizes custom-designed air ends and high-efficiency permanent magnet motors for enhanced operational stability.

Proprietary Air Treatment: Features in-house developed and manufactured after-treatment equipment for superior air purity and consistent quality.

Automated Maintenance: Equipped with automatic drain valves across the system for reduced manual upkeep.

Custom Control System: Employs tailored variable frequency drives and control elements for user-friendly operation and reliable performance.

10000-20000W Laser Cutting Standard Integrated Unit

NO	Technical paramter			
1	Model	TGJ-30VSD-16-500		TGJ-50VSD-16-500
2	Air Delivery Capacity	2.0m³/min	2.6m³/min	3.7m³/min
3	Working pressure	1.6MPa		
4	Transmission Type	Belt transmission		
5	Cooling method	Air cooling		
6	Power	22KW	30KW	37KW
7	Desiccant Air Dryer Capacity / Pressure Dew Point	3.5m³/-40		5.0m³/-40
8	Power supply	380V-220V/50HZ/3-2PH		
9	Air connection diameter	G¾、G½		
10	Rotation speed	2750/2850		2900
11	Capacity of gas tank	560L		
12	Weight	845KG		1050
13	Length × Width × Height	2050*1805*1040		2050X1900X1850
Compatible Laser Power		10000-15000 W	15000-20000W	22000-40000W



Model Advantages:

- Five-in-One Integration:** Unites a permanent magnet variable frequency screw compressor, air treatment system, and twin air receivers into a single, compact unit.
- Multi-Stage Filtration:** Features a multi-stage, high-precision filter configuration, making it ideally suited for high-power laser cutting applications.
- Optimized Logistics & Installation:** The integrated design minimizes secondary pipeline contamination, reduces the overall footprint, and simplifies transportation and on-site installation.
- Premium Core Components:** Utilizes custom-designed airends and high-efficiency permanent magnet motors for superior reliability.
- Proprietary Air Treatment:** Incorporates in-house developed and manufactured after-treatment equipment for exceptional air purity and consistent performance.
- Centralized Automatic Drains:** An integrated automatic drainage system across the unit streamlines maintenance.
- Custom Control System:** Tailored variable frequency drives and control elements ensure user-friendly operation and stable control.
- Scalable Air Storage:** The air receivers include auxiliary ports for easy future capacity expansion.

22000-50000W Laser Cutting Matched Skid-Mounted Integrated Unit

NO	Technical parameter	
1	Model	TGX-50VSD-16-2600
2	Air Delivery Capacity	3.7m³/min
3	Working pressure	1.6MPa
4	Transmission Type	Variable Frequency Direct Drive
5	Cooling method	Air cooling
6	Power	37KW
7	Desiccant Air Dryer Capacity / Pressure Dew Point	5.0m³/-40
8	Power supply	380V/50-60HZ
9	Air connection diameter	G¾、G½
10	Rotation speed	2900
11	Capacity of gas tank	Standard 1200L
12	Weight	1380KG
13	Length × Width × Height	3050X15050X2150
Compatible Laser Power		22000-50000W



Model Advantages:

Fully Integrated Design: Combines a permanent magnet variable frequency screw compressor, air treatment system, and dual 600L air receivers into a single, unified unit.

Enhanced Multi-Stage Filtration: Utilizes a robust, multi-stage, high-precision filter configuration specifically engineered for ultra-high-power laser cutting applications (compatible with 50,000W+ lasers).

Unified Skid-Mounted Layout: All components are mounted on a single, large skid for a compact and integrated piping structure, ensuring easy transportation and rapid on-site installation.

Custom Core Components: Features custom-designed airends paired with high-efficiency permanent magnet motors for superior operational stability.

Proprietary Air Treatment: Incorporates in-house developed and manufactured after-treatment equipment, guaranteeing exceptional air purity and consistent performance.

Automated Maintenance: The entire system is equipped with automatic drains, simplifying upkeep and reducing manual intervention.

Tailored Control System: Employs custom-configured variable frequency drives and control components for user-friendly operation and reliable system control.

Scalable Design: Includes expansion-ready ports, allowing for straightforward future increases in air receiver capacity.

50000-150000W Laser Cutting Integrated Unit

NO	Technical parameter			
1	Model	TGX-60VSD-16-2600	TGX-75VSD-16-2600	TGX-90VSD-16-2600
2	Air Delivery Capacity	5.0m³/min	6.1m³/min	8.2m³/min
3	Working pressure	1.6MPa		
4	Transmission Type	Variable Frequency Direct Drive		
5	Cooling method	Air cooling		
6	Power	45KW	55KW	75KW
7	Desiccant Air Dryer Capacity / Pressure Dew Point	8.5m³/-40		
8	Power supply	380V-220V/50HZ/3-2PH		
9	Air connection diameter	G¾、G½		
10	Rotation speed	3250		
11	Capacity of gas tank	1800L		
12	Weight	1150+750KG	1150+820KG	1150+850KG
13	Length × Width × Height	3000X1600X2150 / 1550*1500*1500		
	Compatible Laser Power	50000-80000W	80000-100000W	100000-150000W



Model Advantages:

Flexible Power Configuration: The permanent magnet variable frequency screw compressor is independently arranged, allowing for adaptable power output.

High-Capacity Air Treatment Skid: The after-treatment system integrates three 600L air receivers, a high-efficiency refrigerated dryer, a desiccant dryer, and a multi-stage enhanced filtration system on a single skid. This configuration is specifically designed for ultra-high-power laser cutting applications (compatible with lasers from 6kW to 150kW and above).

Custom Core Components: Utilizes custom-designed airends and high-efficiency permanent magnet motors for enhanced stability.

Proprietary Air Treatment: Features in-house developed and manufactured after-treatment equipment for superior gas purity and reliability.

Automated Maintenance: The entire system employs automatic drainage for reduced maintenance effort.

Tailored Control System: Incorporates custom-configured frequency converters and control components for user-friendly operation and consistent performance.

Scalable Design: Includes expansion-ready ports for future air receiver capacity increases.

